

Flow direction algorithm based optimal power flow for isolated microgrid integrated with renewable energy sources

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Abstract

This study explores microgrids as small, independent electrical systems that reduce emissions, lower costs, and improve the efficiency and reliability of renewable energy sources (RES). It focuses on solving the optimal power flow (OPF) problem in isolated microgrids to minimize production costs, integrating variable RES like solar photovoltaic systems and wind turbines with a stable small hydropower plant. A novel physics-based Flow Direction Algorithm (FDA), inspired by the D8 hydrological model, is introduced, offering superior precision compared to bio-inspired metaheuristics. The FDA balances global and local search for accurate solutions. Monte Carlo simulation models uncertainties in wind speeds and solar irradiance. Validated on IEEE 33-bus, 69-bus, and 15-bus systems, the FDA outperforms algorithms like Whale Optimization, Ant Lion, Dung Beetle, and Walrus Optimization in efficiency and reliability, advancing microgrid performance.

Keywords

optimal power flow, physics-based optimization, flow direction algorithm, PV plant, wind turbine generators, small hydro power plants, diesel generators

Introduction

Background and motivation

Today, electrical power system operators give priority to address global warming, securing energy, and managing steadily soaring power consumption ([Renewable Energy Agency I, 2018](#)). Many researchers propose microgrids (MGs) as effective solution. MGs are localized energy systems which include interconnected loads, distributed energy resources (DERs), and control systems, and they can operate in two modes: grid-linked or isolated ([Banerji et al., 2013](#)). In grid-linked mode, MGs interact with the main utility grid to efficiently balance supply and demand, while in isolated mode, they function independently, relying solely on local resources and storage systems. Historically, for reliable power sources, isolated MGs used to be dependent on traditional energy sources like diesel generators. However, as idea of sustainability comes into picture, renewable energy sources (RES) like wind and solar started getting integrated to systems which helps in reducing carbon emission footprints and dependency on fuel, but it started to pose challenges that includes intermittency nature of RES which started affecting power stability. This led to the requirement of advanced strategies to maintain the power balance and smooth energy supply.

Isolated MGs are especially encouraged in remote regions where expansion of central grid is economically and technically unfeasible. In India, for example, isolated microgrids have succeeded in areas like Ladakh, Sunderbans, and parts of the Northeast ([Subramony et al., 2017](#)). Specifically, solar-powered microgrids in the Sunderbans, a deltaic region in West Bengal, have significantly improved electrification and amplified local economic activities, such as small-scale

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enterprises. Despite these advantages, isolated MGs face operational challenges due to the intermittency and uncertainty of RES. Keeping power balance between fluctuating power generation and demand, maintaining voltage and frequency stability without grid support, and optimizing RES utilization are critical issues. These challenges highlight the need for advanced Optimal Power Flow (OPF) techniques tailored to the unique dynamics of isolated MGs.

OPF plays an important role in managing power systems as it helps in optimizing some specific objectives like minimization of generating costs, transmission losses, reducing carbon emissions or obtaining optimal load shedding while considering operational constraints. Contrast to traditional metaheuristic methods that draws inspiration from social behaviours (e.g. Particle Swarm Optimization (PSO) and WOA), we are proposing the FDA, a novel physics-based approach rooted in hydrological flow dynamics. It is inspired by the D8 method of drainage basin analysis where FDA offers a unique balance of exploration and exploitation, designed to tackle complex, non-linear constraints of OPF in isolated MGs. In conventional power plants, generation costs increases as output rises from minimum to maximum levels due to rising fuel expenses. Therefore, to address this concern, we integrated solar photovoltaic (PV) units and WTGs into the system. These RESs have low operating costs and produce zero hazardous emissions during power generation, thereby minimizing overall power generation expenses.

Literature review

OPF balances economic feasibility and operational stability in isolated MGs integrating renewable energy sources (RES), yet its complexity necessitates innovative solutions like our proposed FDA. Traditional mathematical methods, such as Linear Programming (LP) (Olofsson et al., 1995), Non-Linear Programming (NLP) (Hau et al., 2000), Quadratic Programming (QP) (Contaxis et al., 1986), Dynamic Programming (DP) (Jadhav, 2018), and Lambda Iteration (Alayande et al., 2019), provide precise solutions for static OPF problems but struggle with the non-linear, stochastic nature of RES, as their computational rigidity limits adaptability to real-time variability. In response, metaheuristic approaches have gained traction for their ability to navigate complex OPF landscapes. For instance, Particle Swarm Optimization (PSO) (Kumar et al., 2018) and Whale Optimization Algorithm (WOA) (Raj and Jaiswal, 2024) efficiently explore solution spaces, while more recent advancements like Moth Swarm Optimization (Mohamed et al., 2017), Artificial Bee Colony (ABC) (Rezaei Adaryani and Karami, 2013), Social Spider Optimization (Nguyen, 2019), Teaching-Learning-Based Optimization (Pandey et al., 2025; Pandey and Mahia, 2023), Krill Herd Algorithm (Roy and Paul, 2015), Dung Beetle Optimization (DBO) (Xue and Shen, 2023), Walrus Optimization Algorithm (WaOA) (Trojovský and Dehghani, 2023), and Ant Lion Optimization (ALO) (Maheshwari and Sood, 2022) offer diverse bio-inspired strategies to avoid local optima. However, these methods, despite their flexibility, often rely on stochastic population updates that lack a physical basis, leading to inconsistent convergence and heuristic tuning challenges (Rajwar et al., 2023). Authors in Hassan et al. (2024) address stochastic OPF in hybrid wind-PV-V2G systems with dynamic load demand, using a hybrid manta ray foraging and artificial hummingbird algorithm (AHMRFO). Their model incorporates Weibull, lognormal, and normal PDFs to simulate RES and EV uncertainties, tested on a modified IEEE 30-bus system.

In isolated MGs, stability and operating costs are critical concerns, prompting specialized studies. Best practices for managing RES-integrated MGs are outlined in Abedini et al. (2016), while coordinated dispatch across cooperative MGs is addressed by Li and Qiu (2016), emphasizing scalability but overlooking dynamic constraints. Authors in Moradi et al. (2016) leverage Harmony Search-Genetic Algorithm (HS-GA) for optimal MG performance, yet neglect long-term RES uncertainty, a gap also evident in (Cagnano et al., 2018)'s cooperative control for remote MG reserves. Grid-linked OPF with energy storage is improved in Levron et al. (2013), but its focus on static scenarios limits relevance to isolated systems. To tackle energy management in remote MGs, a two-step optimization approach—unit commitment (UC) followed by imbalanced 3-phase OPF—is proposed in Olivares et al. (2014), though it assumes simplified frequency stability. Traditional power flow methods falter in such systems due to absent slack buses, prompting alternatives like relaxed bus models with distributed generation (DG) (Kamh and Iravani, 2010; Nikkhajoei and Iravani, 2007), Newton trust zones (Abdelaziz et al., 2013), and modified Newton-Raphson techniques (Mumtaz et al., 2016), which enhance non-linear problem resolution but remain computationally intensive. Linear power flow models, such as backward/forward sweep (Diaz et al., 2016) and its modified version (Hameed et al., 2019), improve efficiency for radial MGs, yet struggle with meshed topologies and dynamic RES fluctuations. Authors in Garcés-Ruiz et al. (2025) develop a quasi-dynamic OPF framework for isolated DC microgrids, combining a fully dynamic model with a convex optimization approach for power sharing. They employ robust constraints and a nodal admittance estimator to handle load uncertainties, validated on a 16-bus system. Yet, their focus on DC systems limits applicability to AC-based isolated MGs with diverse RES like SHPPs.

Uncertainty in RES, a key challenge for OPF, is addressed through robust methods like primal cutting plane decomposition in a two-step framework (Lara et al., 2019), predictor-corrector proximal multipliers for disintegrated OPF (Shi et al., 2015), and semidefinite programming for unbalanced systems (Dall'Anese et al., 2013). Non-linear terms are

simplified into second-order conic constraints in Vergara, Rey, et al. (2019), enabling OPF across grid-connected and islanded modes, while linearization (Vergara, Lopez, et al., 2019) and generalized Bender's decomposition (Raj et al., 2025) reduce computational burden but sacrifice accuracy in dynamic contexts. Multi-objective optimization, as in Dinesh et al. (2014) and Moradi et al. (2014), balances voltage stability, power loss, and DG deployment, yet rarely incorporates SHPPs or long-term planning. While bio-inspired metaheuristics (e.g. PSO, ALO, WOA) mimic natural phenomena like swarm behaviour and predator-prey dynamics, their shared reliance on stochastic movement contrasts sharply with our FDA, which draws from the physical principles of water flow in drainage basins. Using slope-driven dynamics (equation (31)) and adaptive neighbourhood sizing (equation (28)), FDA aligns intuitively with power distribution's flow-like nature, offering enhanced precision and a structured balance of global and local search, unlike the heuristic randomness of its counterparts. Although static OPF dominates current research, dynamic OPF remains underexplored, further underscoring FDA's potential to fill this gap. Authors in Alvarez et al. (2024) propose a dynamic security-constrained AC OPF for microgrids, using differential-algebraic equations and AMPL with IPOPT to ensure stability during islanding, focusing on inverter-based resources and short-term events rather than long-term RES variability. Also Basu (2023a) presents a dynamic OPF for grid-connected multi-microgrids with outages, applying QOFCRCGA to minimize costs across IEEE 33-bus, 15-bus, and 69-bus systems, but its grid-connected scope and bio-inspired approach differ from FDA's physics-based isolated MG focus.

Research gaps

Because of the broad ranges of objective and uses of different function in the problems formulation, any strategy cannot be deemed optimal for solving OPF problem integrating with RESs, according to the literature mentioned above. This encourages the creation and application of innovative metaheuristic algorithms for OPF-related problem solving. Using PDFs to describe the uncertainty of these resources, a lot of studies has been done to find the best power flow result that incorporates WTG and PV generators. The majority of the literature suggests using Weibull and Lognormal PDFs to copy the intermittent behaviour of wind speeds and sun irradiances. However, the models based on probability distribution are not very useful for long-time scheduling or day-ahead OPF problem, so further exploration is needed in given field. The following research gaps has been proposed considering these constraints:

- (1) Much of the existing OPF research integrating RES conducts only brief analyses of uncertainty, relying on static PDF models that offer limited value for long-term or day-ahead planning critical to isolated MGs.
- (2) Traditional OPF methods and even advanced metaheuristics inadequately address RES intermittency, often yielding suboptimal or unreliable solutions due to insufficient modelling of wind speed and solar irradiance variability. Effective, scalable scenario generation remains underdeveloped.
- (3) The inclusion of uncertain RES increases OPF complexity, yet most studies focus narrowly on wind and solar integration, with few exploring alternative resources like small hydropower plants (SHPPs), tidal energy, or energy storage systems, which could enhance system stability.
- (4) Current metaheuristic algorithms predominantly adopt bio-inspired metaphors limiting methodological diversity. Physics-based approaches, such as those inspired by fluid dynamics or terrain flow, remain largely unexamined despite their potential for superior convergence, as FDA demonstrates with its D8-based slope-driven dynamics.

Rationale for choosing the flow direction algorithm

Despite the availability of numerous metaheuristic algorithms, such as Particle Swarm Optimization (PSO), Whale Optimization Algorithm (WOA), Ant Lion Optimizer (ALO), Dung Beetle Optimization (DBO), and Walrus Optimization Algorithm (WaOA), the Flow Direction Algorithm (FDA) is employed in this study due to its unique physics-based approach and superior suitability for the OPF problem in isolated microgrids. Unlike bio-inspired algorithms, which rely on stochastic updates inspired by biological behaviours (e.g. swarm movement or predator-prey dynamics), FDA is rooted in the D8 hydrological model, leveraging physical principles of water flow to navigate the non-linear OPF search space. This physics-based foundation ensures consistent, gradient-driven convergence, reducing the risk of local optima entrapment common in bio-inspired methods. FDA's adaptive neighbourhood radius (equation (28)) balances global exploration and local exploitation, requiring less heuristic tuning than algorithms like PSO or WOA. Furthermore, its slope-driven velocity updates (Equations (30) and (31)) align intuitively with power flow dynamics, offering enhanced precision in handling RES uncertainties and complex constraints. Empirical results (Result and discussion Section) demonstrate FDA's lower costs and tighter cost ranges compared to WOA, ALO, DBO, and WaOA, validated by statistical significance (ANOVA $P < 0.05$). By introducing methodological diversity and addressing the limitations of bio-inspired metaheuristics, FDA is the preferred choice for optimizing isolated microgrid operations.

Contribution and organization of paper

To address the OPF challenges in isolated MGs integrating RES like solar PV units, WTGs, and SHPPs, we propose the Flow Direction Algorithm (FDA), a novel physics-based metaheuristic inspired by the D8 hydrological model (Karami et al., 2021). Unlike bio-inspired metaheuristics such as WOA and ALO, FDA leverages slope-driven dynamics and adaptive neighbourhood sizing to optimize power flow, achieving lower costs and enhanced reliability, as validated on IEEE 33-bus, 69-bus, and 15-bus systems. We employ Monte Carlo simulation (MCS) with a backward reduction technique to generate and refine scenarios, effectively modelling the uncertainty of wind speeds and solar irradiance over a 24-h period, outperforming static probability distribution models. The primary contributions of this paper are as follows:

- **Pioneering a Physics-Based Metaheuristic:** We introduce FDA as the first physics-based approach inspired by the D8 hydrological model for OPF in isolated MGs, addressing the methodological diversity gap by shifting from bio-inspired paradigms and achieving superior convergence, as validated by lower costs.
- **Effective OPF Solution Across Test Systems:** We apply FDA to mitigate OPF challenges in modified IEEE 33-bus, 69-bus, and 15-bus systems, reducing operating costs and demonstrating reliability through tight cost ranges.
- **Advanced Uncertainty Modelling:** We utilize MCS to generate and reduce 2000 scenarios to 10 effectively capturing RES variability and ensuring computational efficiency for mid-to large-scale systems, outperforming static models like Weibull and lognormal PDFs.
- **Comparative Validation:** We benchmark FDA against advanced metaheuristics WOA, ALO, DBO and WaOA, confirming its efficacy with statistically significant lower costs tested by ANNOVA test and faster convergence.

While this study focuses on static OPF, FDA's structured global-local search provides a foundation for future dynamic OPF applications. The paper is organized as follows: Section 2 formulates the OPF problem, Section 3 details the FDA methodology, Section 4 presents case studies and simulation results, and Section 5 summarizes findings and outlines future prospects.

OPF problem formulation

The OPF problem in isolated microgrids (MGs) aims to minimize total operating costs of distributed generation (DG) units—diesel generators, wind turbine generators (WTGs), and solar photovoltaic (PV) units—while treating small hydropower plants (SHPPs) as having negligible operating costs due to their reliance on water flow rather than fuel. This section defines the mathematical framework, justifies assumptions, and derives key equations.

Objective function

The total operating cost C_T over a scheduling period T (e.g. 24 hours) is formulated as the sum of costs from diesel generators, WTGs, and PV units, expressed as

$$C_T = \sum_{t=1}^T \left[\sum_{i=1}^{N_D} C_{DG,i,t}(P_{DGit}) + \sum_{i=1}^{N_W} C_{WTG,i,t}(P_{Wit}) + \sum_{i=1}^{N_{PV}} C_{PV,i,t}(P_{PV,i,t}) \right] \quad (1)$$

Diesel generator cost. It is assumed quadratic due to fuel consumption scaling with power output, justified by standard generator cost models.

$$C_{DG,i,t}(P_{DGit}) = \alpha_i + \beta_i \times P_{DGit} \quad (2)$$

WTG cost. It comprises direct operating cost plus reserve and penalty costs for uncertainty, derived as

$$C_{WTG,i,t}(P_{Wit}) = K_{dwi} \times P_{wit} + Oe_{wit}(P_{wit}) + Ue_{wit}(P_{wit}) \quad (3)$$

The direct cost is $K_{dwi} \times P_{wit}$ (\$).

The reserve costs due to overestimations and the penalty costs because of underestimating of the amount of wind power that may be obtained (Hetzler et al., 2008) are expressed as

$$Oe_{wit}(P_{wit}) = oe_{wi} \times \int_{P_{wit}^{\min}}^{P_{wit}} (P_{wit} - y) \times f_w(y) dy \quad (4)$$

$$Ue_{wit}(P_{wit}) = ue_{wi} \times \int_{P_{wit}}^{P_{wit}^{\max}} (y - P_{wit}) \times f_w(y) dy \quad (5)$$

PV cost. Similarly structured, reflecting solar irradiance uncertainty:

$$C_{PV,i,t}(P_{PV,i,t}) = K_{dPV_i} \times P_{PVit} + Oe_{PVit}(P_{PVit}) + Ue_{PVit}(P_{PVit}) \quad (6)$$

The direct cost is $K_{dPV_i} \times P_{PVit}$ (\$), where K_{dPV_i} is the direct cost coefficient.

In terms of available PV power, reserve costs and penalty costs (Liang and Liao, 2007) are represented as

$$Oe_{PVit}(P_{PVit}) = oe_{PV_i} \times \int_{P_{PVit}^{\min}}^{P_{PVit}} (P_{PVit} - x) \times f_{PV}(x) dx \quad (7)$$

$$Ue_{PVit}(P_{PVit}) = ue_{PV_i} \times \int_{P_{PVit}}^{P_{PVit}^{\max}} (x - P_{PVit}) \times f_{PV}(x) dx \quad (8)$$

Constraints

Both the equality restrictions required from fundamentals rules regulating the distributive systems and inequality constraint implemented from the rating of equipment listed below applied to the objective functions.

Equality constraint. The OPF equation which state that the active and reactive power during given time at every bus of distributive systems must be maintained are known as the equality constraints. The generation of active power should always be equivalent to the summation of active power loss and its demand. Additionally, the aggregate of reactive power generated must always match the total amount of reactive power losses as well as the total amount of reactive power demand.

$$P_{Git} - P_{Lit} - V_{it} \sum_{j=1}^{N_B} V_{jt} \gamma_{ij} \cos(\theta_{ijt} + \delta_{jt} - \delta_{it}) = 0, i \in N_B, t \in T \quad (9)$$

$$Q_{Git} - Q_{Lit} + V_{it} \sum_{j=1}^{N_B} V_{jt} \gamma_{ij} \sin(\theta_{ijt} + \delta_{jt} - \delta_{it}) = 0, i \in N_B, t \in T \quad (10)$$

$$\sum_{i=1}^{N_B} P_{Git} = \sum_{i=1}^{N_B} P_{Lit} + P_{loss_t}, t \in T \quad (11)$$

$$\sum_{i=1}^{N_B} Q_{Git} = \sum_{i=1}^{N_B} Q_{Lit} + Q_{loss_t}, t \in T \quad (12)$$

Inequality constraint. Physical constraints limit bus voltages magnitudes, also the phase angle amid upper and lower bounds

$$V^{\min} \leq V_{it} \leq V^{\max}, i \in N_B, t \in T \quad (13)$$

$$\delta^{\min} \leq \delta_{it} \leq \delta^{\max}, i \in N_B, t \in T \quad (14)$$

Depending on the types of DG being deployed in distributive networks, DG power factor's value lies between upper and lower bounds

$$pf_{DGt}^{\min} \leq pf_{DGt} \leq pf_{DGt}^{\max}, i \in N_d, t \in T \quad (15)$$

Every distribution line's line flow needs to stay inside its limit bounds which is explained as follows:

$$S_{it} \leq S_i^{\max}, i \in N_L, t \in T \quad (16)$$

DG modelling. There are four different kinds of distributive generations where three of these are demonstrated through supplying actual and reactive power to distributive buses in a manner similar to that of SHPPs, WTGs, and diesel generators. Similar to the photovoltaic kind, another one is represented by infusing actual power. Actual active and reactive power generated by the DG should not surpass its capacity restrictions.

$$P_{DGit}^{\min} \leq P_{DGit} \leq P_{DGit}^{\max}, i \in N_d, t \in T \quad (17)$$

$$Q_{DGit}^{\min} \leq Q_{DGit} \leq Q_{DGit}^{\max}, i \in N_d, t \in T \quad (18)$$

Ramp rates limit on diesel generator. During each dispatch period, the ramp rates limit of all committed diesel generator restricts their working ranges.

$$P_{DGit} - P_{DGit(t-1)} \leq UR_i, i \in N_d, t \in T$$

$$P_{DGit(t-1)} - P_{DGit} \leq DR_i, i \in N_d, t \in T \quad (19)$$

Solar power generation modelling. Solar power generation depends on multiple environmental factors, primarily solar irradiance and temperature variations. To model the uncertainty in solar power generation, we have used a lognormal probability distribution function, which effectively characterizes fluctuations in solar irradiance. The solar PV systems' power output is computed based on:

- **Solar Irradiance Forecasts (G_r):** The estimated level of solar radiation available at a given location.
- **Temperature Effects:** The PV panels' output power is influenced by ambient temperature, and we incorporate a temperature coefficient in our equations to account for this variation.

At t hour, resultant solar power from i_{th} Photovoltaic power unit is confirmed by

$$P_{PVit} = P_{PVri} \times \left[1 + \varepsilon_r \times (T_{ref} - T_{ambt}) \times \frac{G_r}{1000} \right], i \in N_{PV}, t \in T \quad (20)$$

Wind power generation modelling. Wind power generation is highly variable due to fluctuations in wind speed. To accurately capture this variability, we have modelled wind power output using a Weibull probability distribution function. The Weibull distribution is widely used for wind energy applications because it effectively represents the stochastic nature of wind speed variations. The power output of a wind turbine at any given time is determined based on the following parameters:

- **Cut-in Speed (V_{in}):** The minimum wind speed at which the turbine begins to generate power.
- **Rated Speed (V_r):** The wind speed at which the turbine produces its maximum power output.
- **Cutout Speed (V_{out}):** The wind speed at which the turbine shuts down to prevent damages.

The operational structures of WTG are used in order to explain the non-linear equation amid power output and wind's velocity. WTG's output power (Shahbazitabar and Abdi, 2018) during given time is shown as

$$P_{wit} = 0, \text{ for } V_{fi} < V_{in} \text{ and } V_{fi} > V_{out}$$

$$P_{wit} = P_{wri} \times \left\{ \frac{(V_{fi} - V_{in})}{(V_{r_i} - V_{in})} \right\}^3, \text{ if } V_{in} \leq V_{fi} < V_{r_i}, j \in N_w, t \in T \quad (21)$$

$$P_{wit} = P_{wri}, \text{ for } V_{r_i} \leq V_{fi} < V_{out}$$

Small Hydro Power Plant (SHPP)

SHPPs is one of the distributed generations (DG) sources within the isolated microgrid. It offers a more stable and predictable power output, reducing the overall uncertainty in microgrid operations. It rely on the controlled release of water from reservoirs, making them dispatchable RES, unlike wind and solar, which are inherently intermittent. The SHPP's generation outputs (Power Generation Operation and Control 3rd edition By Allen J Wood and Bruce F Wollenberg and Gerald B Sheble, n.d.) are determined by the water ejection rate and reservoir storage capacity.

$$Ph_{it} = C_{1i}Vh_{it}^2 + C_{2i}Qh_{it}^2 + C_{3i}Vh_{it}Qh_{it} + C_{4i}Vh_{it} + C_{5i}Qh_{it} + C_{6i}, i \in N_h, t \in T \quad (22)$$

Water equilibrium equation for each SHPP, reservoir water storage limit, and water ejection objectives chosen based on plant and reservoir limitations of the hydro system comprise the hydraulic functional constraints.

$$Vh_i^{\min} \leq Vh_{it} \leq Vh_i^{\max}, i \in N_h, t \in T \quad (23)$$

$$Qh_i^{\min} \leq Qh_{it} \leq Qh_i^{\max}, i \in N_h, t \in T \quad (24)$$

$$Vh_{i(t+1)} = Vh_{it} + Ih_{it} - Qh_{it} - Sh_{it}, i \in N_h, t \in T \quad (25)$$

These limitations are as follows:

- (a) Structural restrictions on the amounts of water that reservoirs may store and discharge,
- (b) The hydro reservoir system's continuity equation

RES scenarios generations

This study integrates three RES: small hydropower plant (SHPP), solar photovoltaic (PV) units, and wind turbine generator (WTG). SHPPs provide stable, dispatchable power due to controlled water release, minimizing uncertainty. In contrast, solar PV and WTG outputs are highly variable due to fluctuations in solar irradiance and wind speed, respectively. To model these uncertainties, we employ Monte Carlo Simulation (MCS) to generate probabilistic scenarios capturing the stochastic nature of these resources over a 24-h scheduling period.

Monte Carlo simulation process. MCS generates 2000 scenarios for solar irradiance and wind speed, reflecting their inherent variability. Solar irradiance is modelled using a lognormal probability distribution function (PDF), $f_{PV}(x)$ which effectively captures the skewed distribution of solar radiation, as described in equation (20). Wind speed is modelled using a Weibull PDF, $f_w(y)$ widely adopted for its ability to represent wind speed variability, as per equation (21). Each scenario is constructed by sampling from these PDFs, incorporating a 10% standard deviation around the mean forecasted values of solar irradiance (G_f) and wind speed (Vf_f). The mean and standard deviation of the generated scenarios approximate a

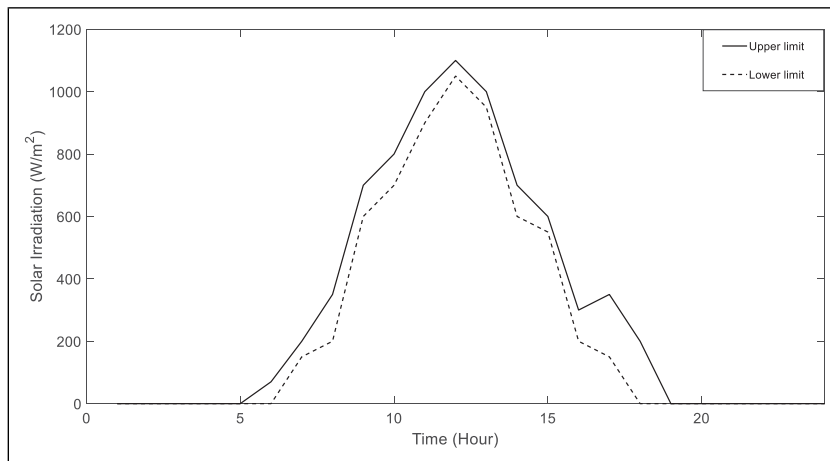


Figure 1. Upper and lower forecasted limit of solar irradiance.

normal distribution, ensuring robust representation of uncertainty. Figures 1 and 2 illustrate the upper and lower forecasted limits for solar irradiance and wind speed, respectively.

Backward reduction technique. To balance computational efficiency with accuracy, the 2000 scenarios are reduced to 10 representative scenarios using the simultaneous backward reduction technique. This method iteratively eliminates scenarios with minimal impact on the overall probability distribution, preserving the statistical properties of the original set. Each of the 10 scenarios is assigned a probability value, enabling the OPF model to account for uncertainty while reducing computational burden (He et al., 2023). This approach outperforms static PDF models, such as standalone Weibull or lognormal distributions, by providing a dynamic, scenario-based representation suitable for day-ahead planning in isolated microgrids.

The power outputs of PV units and WTGs are computed using equations (20) and (21), respectively, based on the reduced scenarios. This methodology ensures that the OPF solution accounts for RES variability, enhancing reliability and cost-effectiveness in isolated microgrid operations.

Flow direction algorithm

The Flow Direction Algorithm (FDA) is a novel physics-based optimization method designed to solve the OPF problem in isolated MGs, drawing inspiration from the D8 hydrological model (Maheshwari et al., 2023). Unlike bio-inspired metaheuristics (e.g. PSO, WOA), FDA leverages physical principles of water flow across a terrain, where elevation represents the cost function C_T , to navigate the complex, non-linear OPF search space. This section expands on FDA's foundation, mechanics, parameter dynamics, convergence characteristics, and tailored application to OPF, providing a comprehensive understanding of its operation and advantages.

FDA principles

D8 (“Eight-Direction” method) to determine the flow direction in a drainage basins serves as foundation for the FDA technique. The D8 approach is a grid-based method that models the flow of water over a terrain represented as a raster (a grid of cells), where each cell has an elevation value. The algorithm assumes that water flows from a given cell to one of its eight neighbouring cells (hence “D8”) based on the steepest downhill slope. This mimics natural drainage patterns in a watershed or drainage basin, where water seeks the path of least resistance toward lower elevations. Inspired by this physical principle, the FDA initializes a population of flows within the search space, representing potential solutions to the OPF. These flows then iteratively move toward lower-cost states, analogous to water flowing downhill, to find optimal results by balancing global exploration and local exploitation, as detailed in subsequent sections. The functioning of algorithm is based on following assumption:

- (1) **Distinct Height and Placement:** Each flow x_i has a unique position (vector of control variables) and cost $C(x_i)$ ensuring a diverse initial population, akin to water starting at varied elevations.

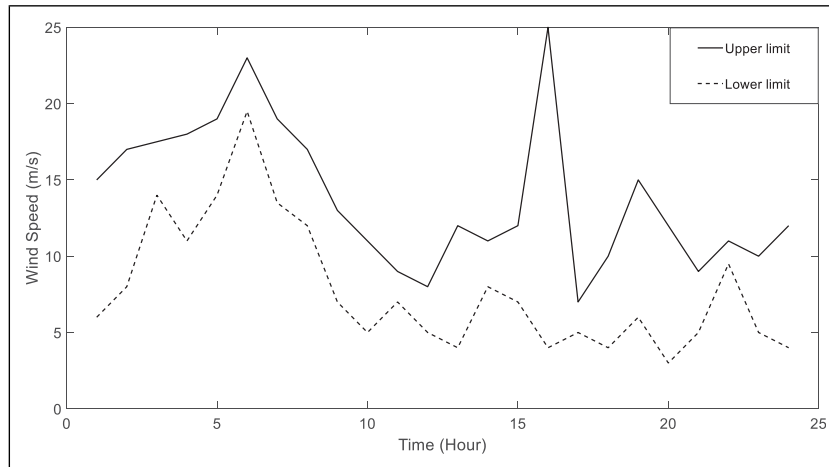


Figure 2. Upper and lower forecasted limit of wind speed.

- (2) **Neighbour Objective Functions:** Each of β neighbours around a flow has a computable cost, enabling slope-based movement, mirroring D8's local gradient assessment.
- (3) **Slope-Velocity Correlation:** Flow velocity V_i is proportional to the cost gradient (slope), reflecting physical flow dynamics where steeper slopes drive faster movement.
- (4) **Lowest Height Movement:** Flows target minimal cost neighbours, analogous to water seeking the basin's lowest point, ensuring optimization toward the global optimum.
- (5) **Outflow as Optimum:** The basin's outflow point represents the ideal solution, aligning with OPF's cost minimization goal.

These assumptions enable FDA to translate hydrological flow into a structured search strategy, distinct from the randomness of bio-inspired methods.

Algorithm mechanics

FDA initializes N_{pop} flows within the OPF search space, defined by control variables such as P_{DGit} , transformer tap settings (TXR), shunt capacitors (Q), and bus voltages (VG), bounded by lb and ub .

The FDA's starting parameters are neighbourhood radius Δ , neighbour count β , and population size N_{pop} .

Initialization. The beginning position of flows is determined by the following expression:

$$Flow_P(i) = lb + rand \times (ub - lb) \quad (26)$$

Here, $rand$ is uniformly distributed varying integer between 0 and 1, $Flow_P(i)$ is the i_{th} Flow point, also lb and ub denotes upper and lower limits placed on controlling variable.

Neighbour generation. Following relationship determines the positions of the neighbourhoods that encircle each flow:

$$Neighbour_P(j) = Flow_P(i) + randn \times \Delta \quad (27)$$

Here, $randn$ is a varying number that is normally distributed and has a standard deviation of one and a mean of zero., and $Neighbour_P(j)$ represents the j_{th} neighbour position.

Radius adaptation. When Δ is small, the search is limited to a small area, but when it is large, it can be conducted across a wide area. While searching over a small area offers an extra accurate global optimum output (local search), searching over a larger region provides a greater range of options, enhancing the possibility to find solutions near to the global optimal value (global search). Therefore, Δ is linearly reduced from higher value to lower value in order for creating an equilibrium amid global and local search. Thus, in order to encourage diversity, Δ is pointed in the following random direction:

$$\Delta = (rand \times P_{rand} - rand \times Flow_P(i)) \times \|Best_P - Flow_P(i)\| \times W \quad (28)$$

Here, P_{rand} denotes the position that is produced at random. With a random value ranging from 0 to infinity, W denotes non-linear weight. The initial component of the connection above indicates that $Flow_P(i)$ is moving in the direction of the P_{rand} . In the second term, the Euclidean distance from $Best_P$ to $Flow_P(i)$ is reduced to zero as the number of iteration grows, bringing it closer to zero. This prevents the local search from working. Equation (29) determines the third term W as follows:

$$W = \left(\left(1 - \frac{current_iter}{max_iter} \right)^{(2 \times randn)} \right) \times \left(\overrightarrow{rand} \times \frac{current_iter}{max_iter} \right) \times \overrightarrow{rand} \quad (29)$$

Here, the current iteration is denoted by $current_iter$, and the max. no. of iteration is represented by max_iter . A uniformly distributed random vector is referred to as a $rand$.

Velocity computation. There is a clear correlation between the slope and flow velocity (V) to neighbour. Hence, flow velocity vector is computed using the relationship displayed in equation (30):

$$V = randn \times S_0 \quad (30)$$

where S_0 is slope vector between neighbour and the flow current point. Following relation provides the i_{th} slope of the flow with respect to the j_{th} neighbour:

$$S_0(i, j, d) = \frac{Flow_fitness(i) - Neighbour_fitness(j)}{\|Flow_P(i, d) - Neighbour_p(j, d)\|} \quad (31)$$

where $Flow_fitness(i)$ and $Neighbour_fitness(j)$ stand for i_{th} Flow's and j_{th} neighbour's objective function values, respectively.

Position update. The dimension of problem is represented with parameter d . The following formula determines the flow's new location:

$$Flow_newP(i) = Flow_P(i) + V \times \frac{Flow_P(i) - Neighbour_P(j)}{\|Flow_P(i) - Neighbour_p(j)\|} \quad (32)$$

where $Flow_newP(i)$ represents new position of i_{th} flow.

Notably, a neighbour's goal function cannot be less than that of flow. By choosing a flow at random, the FDA technique mimics this situation. The flow will go in the same direction if its objective function is smaller than that of the current flow; otherwise, it starts to proceed in direction with the steepest slopes. In these circumstances, flow direction is simulated using the connection displayed in equation (33).

$$\begin{aligned} & \text{if} \\ & Flow_fitness(r) < Flow_fitness(i) \\ & Flow_newP(i) = Flow_P(i) + randn \times (Flow_P(r) - Flow_P(i)) \\ & \text{else} \\ & Flow_newP(i) = Flow_P(i) + 2randn \times (Best_P - Flow_P(i)) \\ & \text{end} \end{aligned} \quad (33)$$

Here, r is a random integer.

Convergence and computational aspects

FDA's convergence stems from Δ_t 's decay and slope-driven updates, reducing search scope while directing flows to cost minima. Unlike PSO's heuristic tuning, FDA's physics-based logic ensures consistent progression, validated by tight cost. Computational complexity is $O(N_{pop} \cdot \beta \cdot T_{max})$ comparable to metaheuristics but optimized by Δ_t 's adaptability, reducing evaluations in later iterations.

FDA algorithm for OPF

The suggested algorithm for handling OPF problems is explained in full in the steps that follow. Step 1: equation (26) is used to randomly initialize the flow matrix (Flow P), and N_{pop} determines the size of flow matrix. Every single one of Flows offers a possible remedy for OPF issues. It includes randomly generated control variables including transformer tap settings (T_{XR}), shunt capacitors (Q), and active power output of diesel generators bus (P_D) disregarding slack bus and voltages magnitude (V_G). The FDA algorithm's whole search space for every single flow is shown as under:

$$Flow_P = \begin{bmatrix} P_{G,2}^1 \dots P_{G,NG}^1 & V_{G,1}^1 \dots V_{G,NG}^1 & T_{XR,1}^1 \dots T_{XR,NT}^1 & Q_1^1 \dots Q_{NSVC}^1 \\ P_{G,2}^i \dots P_{G,NG}^i & V_{G,1}^i \dots V_{G,NG}^i & T_{XR,1}^i \dots T_{XR,NT}^i & Q_1^i \dots Q_{NSVC}^i \\ P_{G,2}^{N_{POP}} \dots P_{G,NG}^{N_{POP}} & V_{G,1}^{N_{POP}} \dots V_{G,NG}^{N_{POP}} & T_{XR,1}^{N_{POP}} \dots T_{XR,NT}^{N_{POP}} & Q_1^{N_{POP}} \dots Q_{NSVC}^{N_{POP}} \\ . & . & . & . \end{bmatrix} \quad (34)$$

Step 2: The following is the formulation of the fitness function that incorporates the penalty function:

$$F = f_{obj} + \left[\lambda_p (P_{G,1} - P_{G,1}^{lim})^2 + \lambda_Q \sum_{i=1}^{NG} (Q_{G,i} - Q_{G,i}^{lim})^2 + \lambda_V \sum_{i=1}^{NL} (V_L - V_{L,i}^{lim})^2 + \lambda_S \sum_{i=1}^{NTL} (S_i - S_i^{lim})^2 \right] \quad (35)$$

Here, $P_{G,1}$ stands for slack bus active power, Q_G for diesel generator bus reactive power, V_L for the load bus voltages, and S for line flow. The letters $P_{G,1}^{\text{lim}}$, $Q_{G,i}^{\text{lim}}$, $V_{L,i}^{\text{lim}}$, and S_i^{lim} stand for the min. and max. limitation of the active power of slack bus, the reactive power of generating buses, load bus voltages, and line flows. The total no. of transmission lines, load bus, and generator buses are denoted by the letters NG , NL , and NTL , respectively. The weighted factors for each constraint are λ_p , λ_Q , λ_V , and λ_S . In order to avoid the iterative process producing impractical Flows, this study chooses comparatively high weight factor values.

Step 3: Sort the individual flows based on their fitness to find the greatest one.

Step 4: Using equation (27), every individual in flow is given a group of neighbours under given radius, and based on their level of fitness, the finest neighbour is selected.

Step 5: Flow moves closer to the neighbour if fitness of neighbouring individual is greater than the Flow's, and equations (30) and (32), respectively, are employed to modify the flow's position and velocities. On the other hand, flow will keep going in same direction, and equation (33) will be used to update its location if its fitness is greater than that of its neighbour.

Step 6: If the new flows' fitness surpasses that of the old Flows, update their position and fitness.

Step 7: Any violation of a control variable cv_i is dealt with as follows:

$$\begin{aligned} cv_i &= cv_{\max} \text{ if } cv_i > cv_{\max} \\ &= cv_{\min} \text{ if } cv_i < cv_{\min} \end{aligned} \quad (36)$$

Step 8: After achieving the max no. of iteration, halt the operation and print the ideal schedule; if not, move on to Step 3.

Innovative aspects of FDA

The FDA distinguishes itself from existing OPF optimization techniques through several innovative features.

- Its physics-based foundation, rooted in the D8 flow direction method, contrasts with the bio-inspired paradigms dominating the field, providing a novel lens for search space exploration.
- The adaptive neighbourhood radius (Δ)—which linearly decreases from a global to local scope (equation (28))—offers a structured yet flexible balance between exploration and exploitation, unlike the often-heuristic tuning in PSO or WOA.
- The slope-driven velocity update (equations (30) and (31)) mirrors physical gradients, ensuring directional convergence toward optimal solutions with a clarity absent in swarm-based randomness.

These attributes collectively position FDA as a methodological advancement, particularly suited to the multi-dimensional, constrained nature of microgrid OPF. The FDA algorithm's flowchart is displayed in Figure 3.

Result and discussion

This study evaluates the Flow Direction Algorithm (FDA) for OPF in isolated MGs integrating RESs—solar PV units, WTGs, and SHPPs—alongside diesel generators. Simulations were conducted on a personal computer (Intel Core i5 11th Gen, 16 GB RAM) using MATLAB R2023a, testing FDA against WOA, ALO, DBO, and WaOA on modified IEEE 33-bus, 69-bus, and 15-bus systems. Figure 4 shows the temperatures throughout the day.

Simulation setup and metrics

Each test system incorporates diesel generators, WTGs, PV units, and an SHPP, with parameters from Tables 1, 2, and 3. RES uncertainties (wind speed V_{ft} , solar irradiance G_r) were modelled via MCS with 2000 scenarios reduced to 10, using Weibull and lognormal PDFs. Load demand varies hourly, with temperature effects on PV output. All 3-test system solar PV units have a maximum power generation capability of 300 kW. All RES related data has been taken from Basu (2023b).

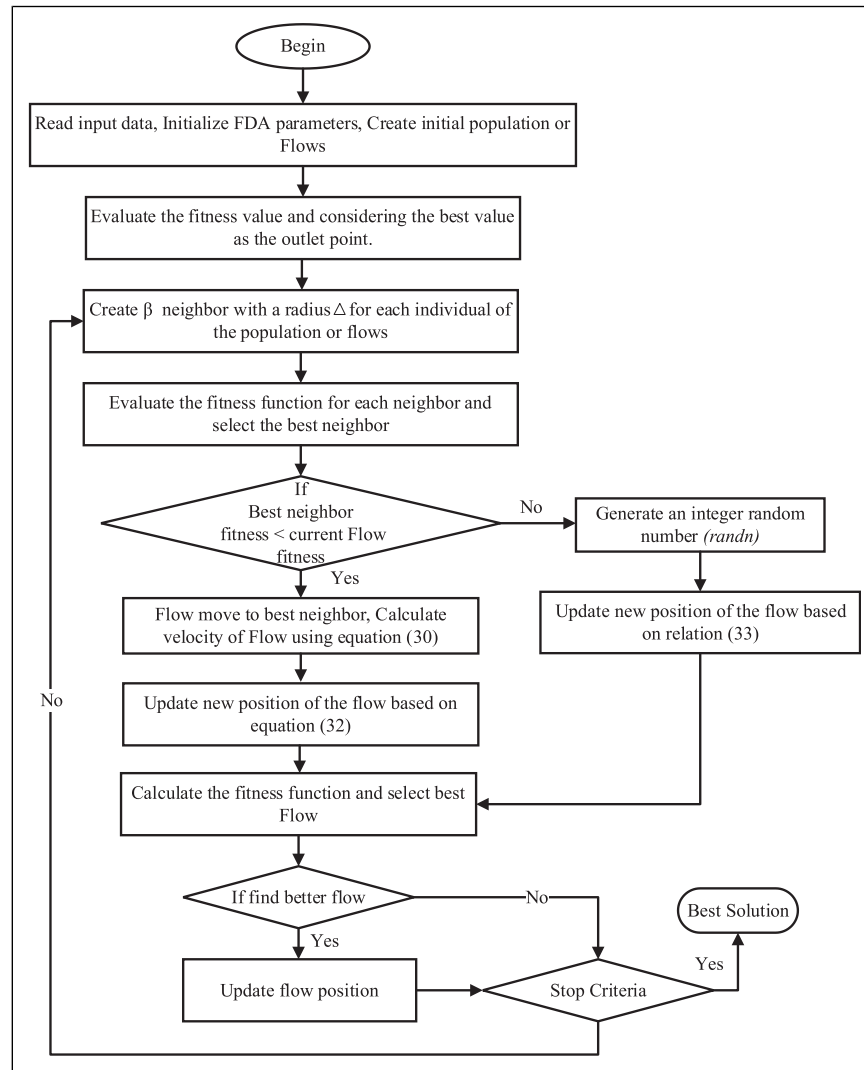


Figure 3. FDA algorithm flow chart.

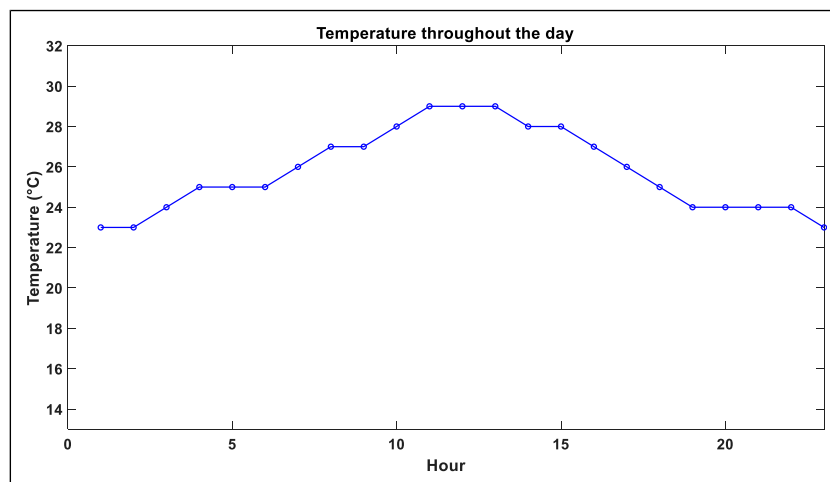


Figure 4. Temperature throughout the day.

Table 1. Diesel generators parameter connected IEEE 33-bus system.

Generators	Buses	Active power (MW)	Reactive power (MW)	UR (MW/h)	DR (MW/h)
Diesel 1	1	1.5	1.2169	0.3	0.3
Diesel 2	6	1.2	0.8186	0.1	0.1
Diesel 3	13	1	0.8176	0.2	0.2
SHPP	21	4	3.5		

Table 2. Diesel generators parameter connected IEEE 69-bus systems.

Generators	Buses	Active power (MW)	Reactive power (MW)	UR (MW/h)	DR (MW/h)
Diesel 1	1	1.5	1.2169	0.2	0.2
Diesel 2	6	1.0	0.8186	0.1	0.1
Diesel 3	25	1.2	1.0176	0.1	0.1
Diesel 4	45	1.5	1.2180	0.1	0.1
SHPP	61	4	3.5		

Table 3. Diesel generators parameter connected IEEE 15-bus systems.

Generators	Buses	Active power (MW)	Reactive power (MW)	UR (MW/h)	DR (MW/h)
Diesel 1	1	0.9	0.9170	0.3	0.3
Diesel 2	4	0.8	0.8186	0.3	0.3
SHPP	3	0.4	0.2		

IEEE 33-bus system

The IEEE 33-bus system (12.66 kV, 32 branches) and the main feeder makes up the initial system (Moradi et al., 2014). It includes three diesel generators (buses 1, 6, 13), one SHPP (bus 23), four PV units (buses 9, 11, 21, 22) and two WTGs (buses 27, 29). $P_{wr1} = 600$ KW and $P_{wr2} = 400$ KW are the ratings for WTGs. Table 1 has displayed their parameters. SHPP data is taken from Basu (2023b). $v_{in} = 5$ m/sec, $v_o = 25$ m/s, and $v_r = 15$ m/s are cut in, cutout, and rated wind speed. For each WTG, the selected direct costs coeff. (K_{dw}), reserve costs (oe_w), and penalty costs (ue_w) are 0.07, 0.02, and 0.01 correspondingly. The active and reactive power loads of several bus for a 24-h period are listed in Table 4.

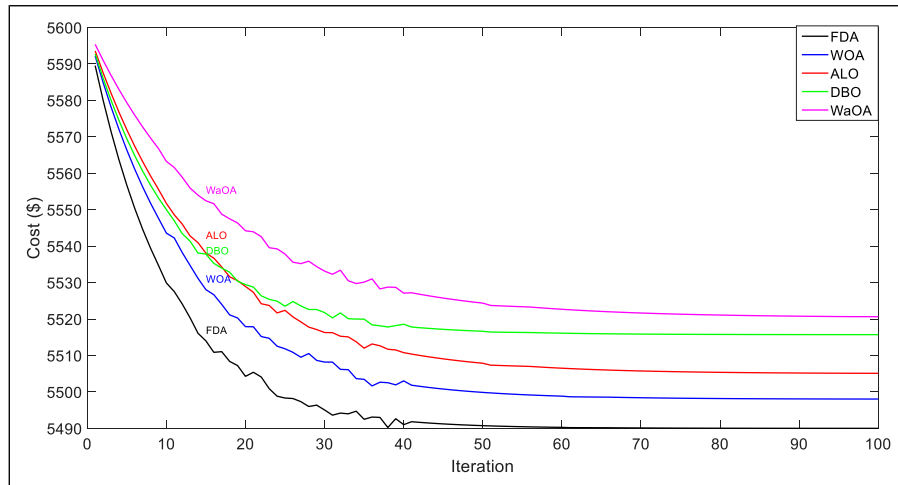
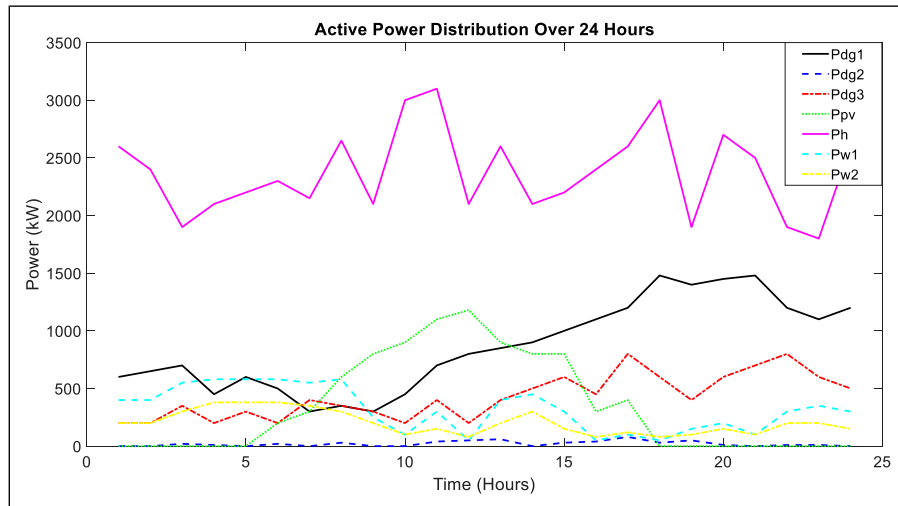
For all five methods, the maximum iteration number is set at 100. Table 5 provides a summary of the best, average, and worst costs across 30 runs for solution produced by FDA, WOA, ALO, DBO and WaOA methods, respectively. The cost

Table 4. Active power load and reactive power loads on various bus.

Hour (t)	Active power load (P_{Li} , KW)	Reactive power load (Q_{Li} , KVAR)	Multiplication factor applied
1–2	P_{Li}	Q_{Li}	Base load (1×)
3–4	$1.05 \times P_{Li}$	$1.05 \times Q_{Li}$	5% increase
5–6	$1.1 \times P_{Li}$	$1.1 \times Q_{Li}$	10% increase
7–8	$1.15 \times P_{Li}$	$1.15 \times Q_{Li}$	15% increase
9–10	$1.2 \times P_{Li}$	$1.2 \times Q_{Li}$	20% increase
11–12	$1.25 \times P_{Li}$	$1.25 \times Q_{Li}$	25% increase
13–14	$1.3 \times P_{Li}$	$1.3 \times Q_{Li}$	30% increase
15–16	$1.35 \times P_{Li}$	$1.35 \times Q_{Li}$	35% increase
17–18	$1.25 \times P_{Li}$	$1.25 \times Q_{Li}$	25% increase
19–20	$1.2 \times P_{Li}$	$1.2 \times Q_{Li}$	20% increase
21–22	$1.15 \times P_{Li}$	$1.15 \times Q_{Li}$	15% increase
23–24	$1.1 \times P_{Li}$	$1.1 \times Q_{Li}$	10% increase

Table 5. Performance comparison for the IEEE 33-bus systems.

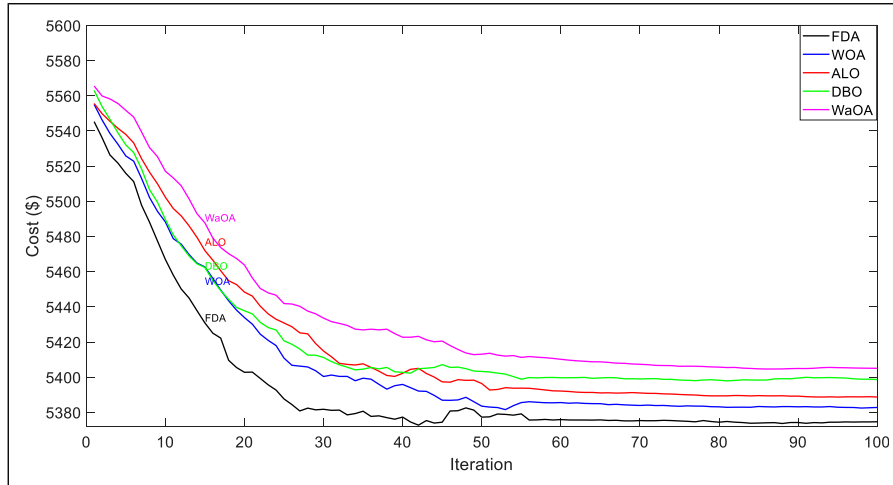
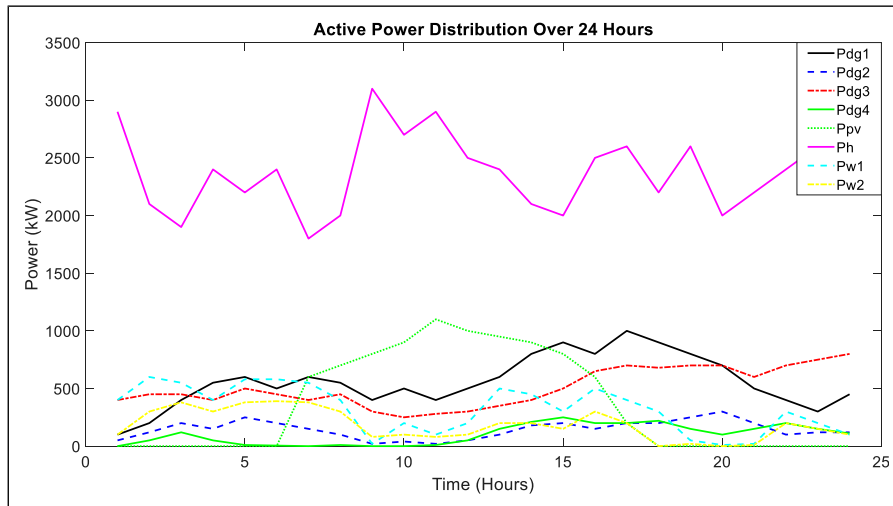
Algorithm	Best costs	Average costs	Worst costs	Time (sec)
FDA	5490.6	5495.6	5500.7	60.2934
WOA	5498.5	5503.5	5508.5	42.6532
ALO	5505.8	5512.05	5518.3	50.1296
DBO	5515.7	5522	5528.3	55.9875
WaOA	5520.4	5531.15	5541.9	69.2897

**Figure 5.** Costs convergence characteristic for IEEE 33 bus systems.**Figure 6.** Active power generation found from FDA for IEEE 33-bus systems.

obtained from the FDA is 5490.6 \$, WOA is 5498.5 \$, ALO is 5505.8 \$, DBO is 5515.7\$ and WaOA is 5520.4\$. The FDA produced 0.144%, 0.277%, 0.457% and 0.543% lower than the WOA, ALO, DBO and WaOA. Figure 5 compares the FDA's convergence curves with those of other applied approaches. The outcomes show how accurate and validated the suggested algorithm is when compared to alternative metaheuristic methods. Figure 6 displays active power production found from FDA for IEEE 33-bus systems. Modified IEEE 33-bus system is shown in Figure A11.

Table 6. Performance comparison for the IEEE 69-bus systems.

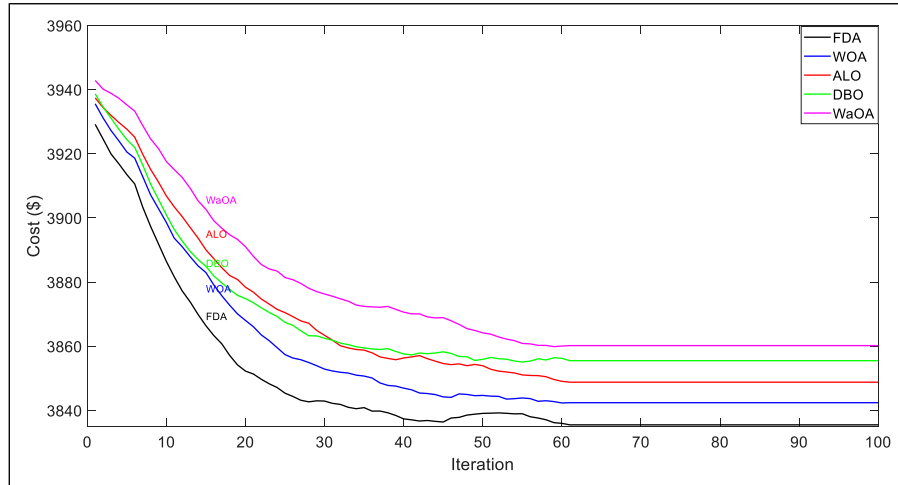
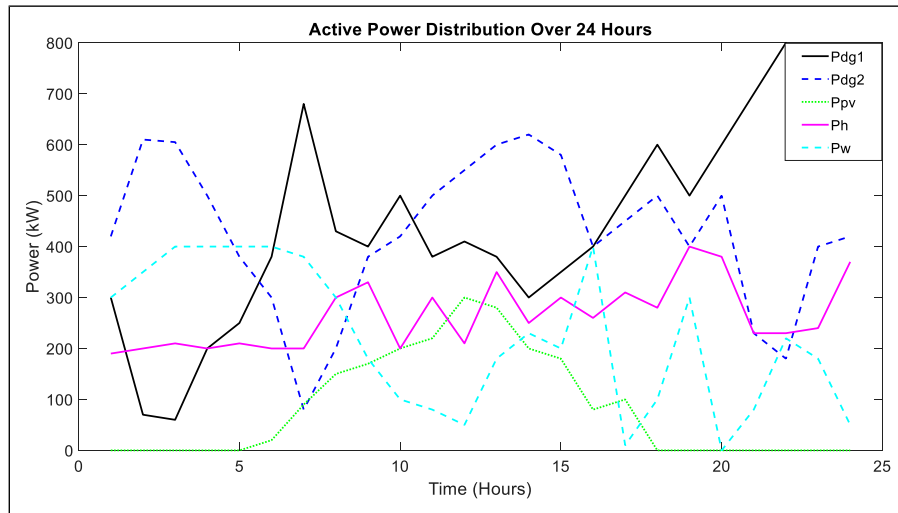
Algorithm	Best costs	Average costs	Worst costs	Time (sec)
FDA	5375.6	5380.9	5386.2	560.8754
WOA	5383.4	5389.55	5395.7	480.2981
ALO	5388.8	5394.15	5399.5	450.6542
DBO	5398.6	5405.6	5412.6	470.9762
WaOA	5404.2	5412.25	5420.3	480.1279

**Figure 7.** Costs convergence characteristic for IEEE 69-bus systems.**Figure 8.** Active power generation obtained from FDA for IEEE 69-bus system.

The base active along with reactive power loads of the i_{th} buses are shown by P_{Li} and Q_{Li} . The multiplication factors indicate how the power demand increases during different hours of the day due to peak consumption periods. For IEEE 33-bus systems, $i = 1, 2, \dots, 33$ (P_{Li} and Q_{Li} are 3715 KW, 2300KVAR); for the IEEE 69-bus systems, $i = 1, 2, \dots, 69$ (P_{Li} and Q_{Li} are 3802.19 KW, 2694.60KVAR), $i = 1, 2, \dots, 15$ (P_{Li} and Q_{Li} are 1322 KW, 820KVAR) for 15-bus systems.

Table 7. Performance comparison for the IEEE 15-bus systems.

Algorithm	Best costs	Average costs	Worst costs	Time
FDA	3835.5	3840.85	3846.2	30.3245
WOA	3842.4	3847.95	3853.5	19.5899
ALO	3848.8	3854.30	3859.8	22.1274
DBO	3855.5	3860.1	3864.7	35.0192
WaOA	3860.2	3866.4	3872.6	38.1439

**Figure 9.** Costs convergence characteristic for IEEE 15-bus systems.**Figure 10.** Active power generation found by FDA for IEEE 15-bus systems.

IEEE 69-bus systems

The IEEE 69-bus system (12.66 kV, 68 branches) is implemented (Moradi et al., 2014). It features four diesel generators (buses 1, 6, 25, 45), one SHPP (bus 61), four PV units (buses 9, 11, 21, 22), and two WTGs (buses 27, 29). Table 2 has displayed their parameters. WTG have a $P_{wr1} = 600$ KW and $P_{wr2} = 400$ KW rating. $v_{in} = 5$ m/sec, $v_o = 25$ m/s, and $v_r = 15$ m/s are the cut in, cutout, and rated wind speed. For each WTG, selected direct costs coeff. (K_{dw}), reserve costs (o_{ew}) and

Table 8. Results of statistical ANOVA test.

System	Source of variations	SS	df	MS	F	P-value
IEEE 33 bus	Between group	64,771.65	4	16,192.91	11.3230	5.0419E-08
	Within group	207,362.25	145	1430.08		
	Total	272,133.9	149			
IEEE 69 bus	Between group	2.05 E+08	4	51,258,465	18.1279	4.29E-12
	Within group	4.10 E+08	145	2,827,586		
	Total	6.15 E+08	149			

penalty costs (u_{ew}) are 0.07, 0.02, and 0.01 correspondingly. Table 4 shows the data of active and reactive power loads of several bus for 24-h period.

For all five methods, the maximum iteration number is set at 100. Table 6 provides a summary of the best, average, and worst costs across 30 runs for solution produced by FDA, WOA, ALO, DBO and WaOA methods, respectively. The cost obtained from the FDA is 5375.6 \$, WOA is 5383.4 \$, ALO is 5388.8 \$, DBO is 5398.6\$ and WaOA is 5404.2\$. The FDA produced 0.145%, 0.246%, 0.428% and 0.532% lower than the WOA, ALO, DBO and WaOA. Figure 7 compares the FDA's convergence curves with those of other applied approaches. The outcomes show how accurate and validated the suggested algorithm is when compared to alternative metaheuristic methods. Figure 8 displays active power production found from FDA for IEEE 69-bus systems. Modified IEEE 69-bus system is shown in Figure A12.

IEEE 15-bus systems

The IEEE 15-bus system (11 kV) is the third system (Das et al., 1995). It has two diesel generators (buses 1, 4), one SHPP (bus 3), one PV unit (bus 9), and one WTG (bus 14). Table 3 displays their parameters. WTG has a $P_{wl} = 400$ KW rating. $v_{in} = 5$ m/sec, $v_o = 25$ m/s, and $v_r = 15$ m/s are the cut in, cutout, and rated wind speed, respectively. For each WTG, the selected direct cost coefficient (K_{dw}), reserve costs (o_{ew}), and penalty costs (o_{ew}) are 0.07, 0.02, and 0.01 correspondingly. Table 4 shows the active and reactive power loads of several bus over a 24-h period.

For all five methods, the maximum iteration number is set at 100. Table 7 provides a summary of the best, average, and worst costs across 30 runs for solution produced by FDA, WOA, ALO, DBO and WaOA methods, respectively. The cost obtained from the FDA is 3835.5 \$, WOA is 3842.4 \$, ALO is 3848.8 \$, DBO is 3855.5\$ and WaOA is 3860.2\$. The FDA produced 0.180%, 0.347%, 0.521% and 0.644% lower than the WOA, ALO, DBO and WaOA. Figure 9 compares the FDA's convergence curves with those of other applied approaches. The outcomes show how accurate and validated the suggested algorithm is when compared to alternative metaheuristic methods. Figure 10 displays active power production found from FDA for IEEE 15-bus systems. Modified IEEE 15-bus system is shown in Figure A13.

The superior performance of FDA for IEEE 33-bus, 69-bus, and 15-bus systems can be attributed to its physics-inspired design. Unlike WOA and ALO, which rely on randomized population updates, FDA's slope-based movement (equation (31)) ensures consistent progression toward lower-cost solutions, while its adaptive Δ (equation (28)) prevents premature convergence—a common pitfall in DBO and WaOA. This structured search strategy yields tighter cost ranges for IEEE 33-bus, 69-bus and 15-bus systems compared to competitors, underscoring its reliability and precision.

To compare several solution approaches and validate the effectiveness of the suggested algorithm, one-way ANOVA, or “analysis of variance,” is carried out. The ANOVA test is carried out to ascertain whether statistically significant differences are revealed by the computed mean of the different solution approaches for each run. A 5% significance level is taken into account in this study; if the P-value is less than 5%, the performance of the recommended algorithm deviates noticeably from alternative approaches. The results of a one-way ANOVA on all algorithms are shown in Table 8, where it is noted that the P-value is significantly less than 0.05, supporting the idea that the recommended method is statistically different from the others.

Conclusion

This study presents the FDA-based OPF for an isolated MG which incorporates RESs. MC simulation is used to generate scenarios in order to overcome the uncertainty of RESs. DGs, SHPP facilities, wind turbine generator, and PV plants make up a microgrid. The Flow direction algorithm was used to solve these issues, and the results were compared with WOA and ALO. For validation, three test system the IEEE 33 bus systems, IEEE 69 bus systems and the IEEE 15 bus systems were employed. On three test system, the Flow direction algorithm yields the lowest cost of the five methods. Dynamic OPF for

isolated and grid-linked multi-MG systems is the future focus of this study. Beyond dynamic OPF, future work could explore FDA's applicability to other power system challenges such as integrating FDA with placement optimization, possibly using multi-objective or hybrid methods to co-optimize location and operational dispatch, multi-objective optimization (e.g. cost vs. emissions) or real-time control of grid-connected microgrids. Additionally, hybridizing FDA with machine learning techniques—for example, using neural networks to predict optimal adaptive neighbourhood radius, Δ values—could further enhance its adaptability, broadening its impact across energy optimization domains.

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Ankur Maheshwari: Supervision, Project administration.

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Data Availability Statement

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix

Nomenclature

Min	Minimum value
Max	Maximum value
Ref	Reference
WTG	Wind turbine generator
C_T	Cost function (\$)
N_D, N_W, N_{PV}	Number of diesel generators, WTGs, and PV units, respectively
$P_{DGit}, P_{wit}, P_{PVit}$	Active power outputs (MW) at time t for the i_{th} diesel generator, WTG, and PV unit, respectively
$C_{DGit}, C_{WTGit}, C_{PVit}$	Cost functions (\$/h) for diesel, WTG and PV unit type at time t
α_i and β_i	Cost coefficient (\$/MW ² h) and maintenance costs (\$/MWh).
K_{dPVi}	Direct costs coefficient for the i_{th} PV (\$/MW)
K_{dwi}	Direct cost coefficient for the i_{th} WTG (\$/MW)
G_r	PV irradiation forecasts (W/m ²)
$f_w(y)$	Weibull probability distribution fn. of wind power y
$f_{PV}(x)$	Lognormal probability distribution fn. of PV power x
I_{hit}	Inflow rate of i_{th} reservoirs at time t (hm ³ /h) N no. of unit
$Oe_{PVit}(P_{PVit})$	Reserve costs function due to overestimation of the i_{th} solar plant at time t
$Oe_{wit}(P_{wit})$	Reserve costs function due to overestimation of i_{th} WTG at time t
$P_{DGit}^{min}, P_{DGit}^{max}$	Min and max active power generation of the i_{th} diesel generator (MW)
P_{hit}	Power produced of i_{th} SHPP at time t (MW)
h_i^{min}, P_{hi}^{max}	Min. and max. power production limit of i_{th} SHPP (MW)
P_{Lit}	Active power demand at i_{th} bus at hour t (MW)
P_{loss_t}	Total active power loss at hour t (MW)
P_{PVri}	Rated power output of i_{th} PV unit (MW)
P_{wri}	Rated power output of i_{th} WTG (MW)
pf_{DGit}	Power factor of i_{th} diesel generators at hour t
$pf_{DGit}^{min}, pf_{DGit}^{max}$	Min. and max. power factor of i_{th} diesel generator
Q_{Git}	Reactive power generation of i_{th} bus at hour t (MVar)
Q_{DGit}	Reactive power generation at i_{th} diesel generators at hour t (MVar)
$Q_{DGit}^{min}, Q_{DGit}^{max}$	Min. and max. reactive power generation of the i_{th} diesel generator (MVar)
Q_{Lit}	Reactive power demand at i_{th} bus at hour t (MVar)

Q_{loss_t}	Total reactive power loss at hour t (MVar)
Q_{hit}	Water ejection rate of i_{th} reservoir at time t (hm^3/h)
$Q_{hi}^{\min}, Q_{hi}^{\max}$	Max and min water ejection rate of i_{th} reservoir (hm^3/h)
UR_i, DR_i	Ramp-up and ramp-down rates limit at i_{th} diesel generators (MW/h)
$U_{epv_{it}}(P_{pv_{it}})$	Penalty costs function due to underestimation of the i_{th} solar plants at time t
ue_{pv_i}, oe_{pv_i}	Penalty costs and reserve costs for the i_{th} PV plant (\$/MW)
$U_{ew_{it}}(P_{w_{it}})$	Penalty costs function due to underestimation of i_{th} WTG at time t
ue_{w_i}, oe_{w_i}	Penalty costs and reserve costs for the i_{th} WTG (\$/MW)
V_{in}	Cut in wind speed (m/s)
V_{out}	Cutout wind speed (m/s)
V_r	Rated wind speed (m/s)
V_{f_t}	Forecasted wind speed at time t (m/s)
$V_{h_{it}}$	Storage capacity of i_{th} reservoir at hour t (hm^3)
$V_{h_i}^{\min}, V_{h_i}^{\max}$	Min. and max. storage capacity of i_{th} reservoir (hm^3)
$V_{h_{i0}}$	Beginning storage capacity of i_{th} reservoir (hm^3)
$V_{h_{iT}}$	Finishing storage capacity of i_{th} reservoir (hm^3)
Y_{ij}	Magnitude of the ij_{th} element of bus admittance bus matrix (mho)
V_{it}	Magnitude of i_{th} bus voltages at hour t (KV)
Sh_{it}	Spillage of i_{th} reservoir at time t (hm^3/h)
$S_i^{\max}$	Max power flow of i_{th} line (MVA)
S_{it}	Power flow of i_{th} line at hour t (MVA)
T_{ref}, T_{ambt}	Ref. and ambient temperature (C)
δ_{it}	Phase angle of i_{th} bus voltage at hour t
t, T	Time index and scheduling period
α_i, β_i	Cost coefficient of i_{th} diesel generator (\$, \$/MW)
ε_r	Temperature coefficient

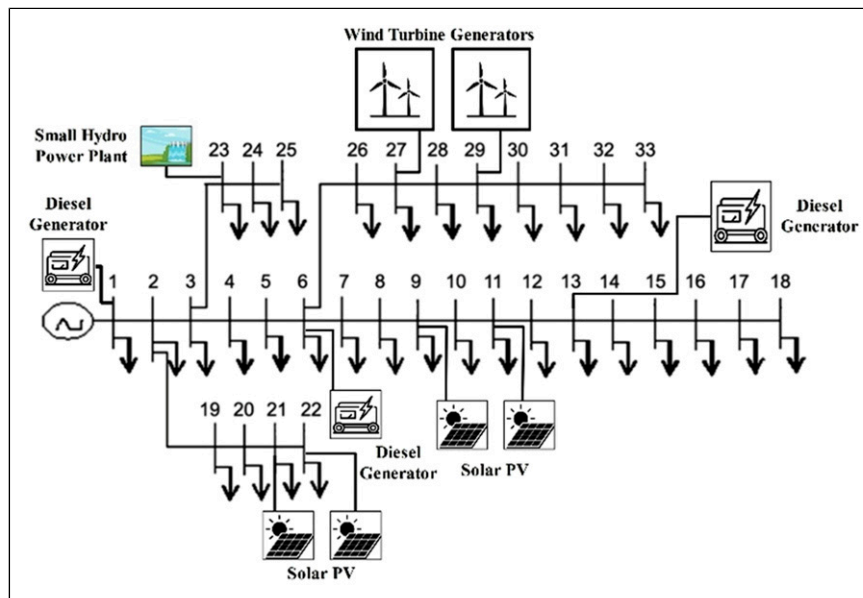


Figure A11. Modified IEEE 33-bus system.

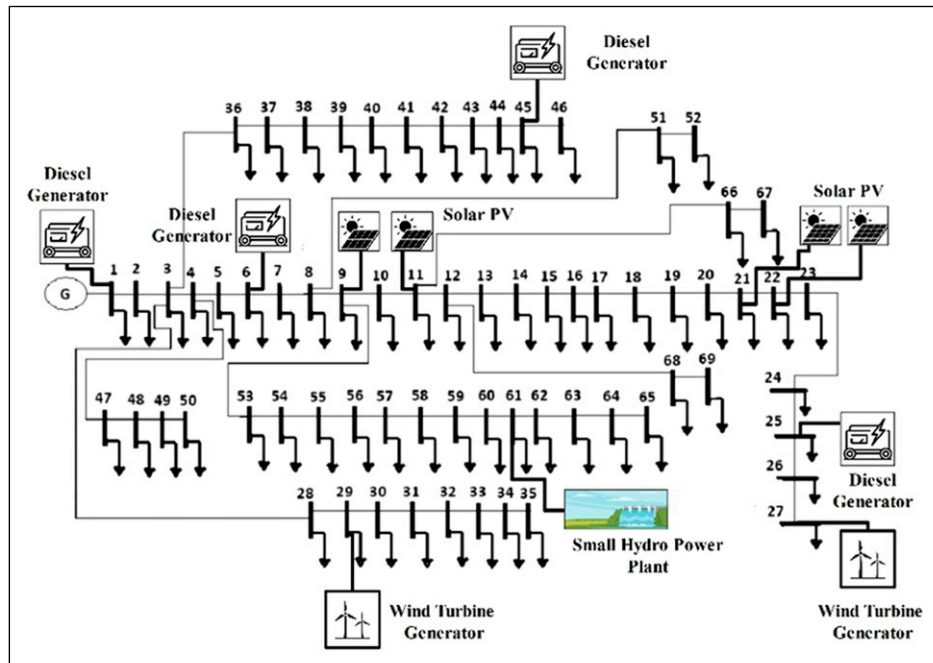


Figure A12. Modified IEEE 69-bus system.

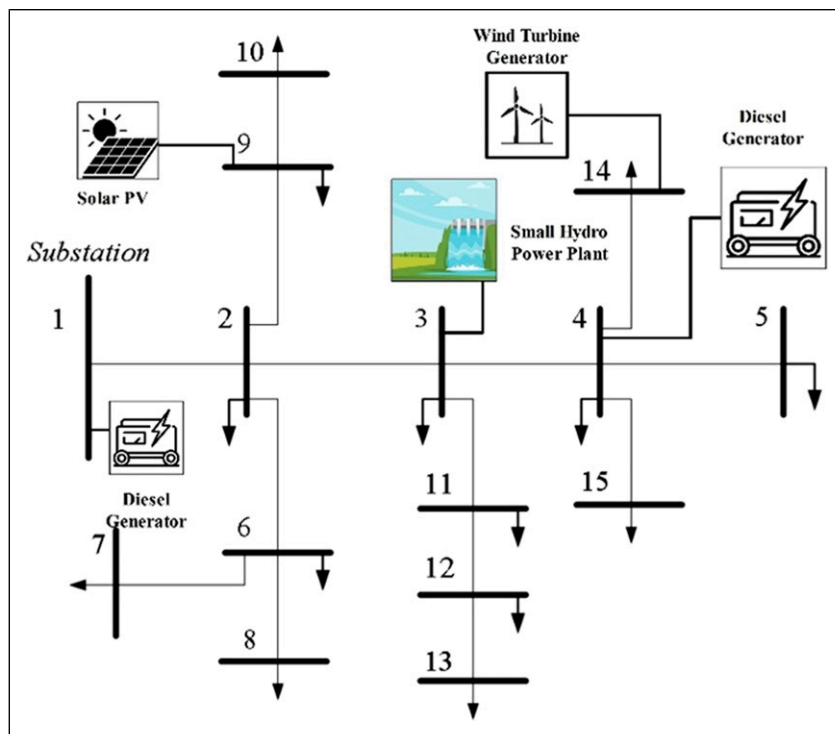


Figure A13. Modified IEEE 15-bus system.